Optimal Growth

The economic problem

The planner chooses consumption and next-period capital:

$$\max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad k_{t+1} = F(k_t) - c_t, \quad c_t > 0,\ k_{t+1} \geq 0.$$

Primitives

$$\beta = 0.96, \quad \gamma = -2, \quad \alpha = \tfrac{1}{4}$$

$$u(c) = -\frac{1}{c}, \qquad F(k) = k + \frac{1}{6}k^{1/4}.$$

Euler equation

$$c(k) = \frac{c(g(k))}{\sqrt{\beta F'(g(k))}}, \quad g(k) = F(k) - c(k),$$

$$F'(k) = 1 + \frac{1}{24}k^{-3/4}.$$

Goal: compute $c(k)$ on $D = [0.8, 1.2]$. Steady state: $k^* = 1$, $c^* = 1/6$.

Approximate the consumption policy

$$c_a(k) = \sum_{j=0}^{n-1} a_j \phi_j(k), \qquad x(k) = 5(k-1) \in [-1, 1].$$

Basis	First functions / representation	Main feature
Chebyshev	$1,\ x,\ 2x^2 - 1, \ldots$	Global polynomials
Legendre	$1,\ x,\ (3x^2 - 1)/2, \ldots$	Same polynomial space
Monomials	$1,\ x,\ x^2, \ldots$	Conditioning worsens with degree
Linear hats	Piecewise-linear interpolation	Local support; a_j are knot values

Compare $n \in \{5, 9, 13, 21\}$ coefficients.

Projection: define the residual conditions

$$R(k;a) = \frac{1}{c^*}\left[c_a(k) - \frac{c_a(g_a(k))}{\sqrt{\beta F'(g_a(k))}}\right], \qquad g_a(k) = F(k) - c_a(k).$$

Collocation

$$R(k_i;a) = 0, \qquad i = 1,\dots,n.$$

Zero residual at selected nodes.

Galerkin

$$\int_D R(k;a)\phi_j(k)\,dk = 0.$$

Residual orthogonal to basis functions.

Least squares

$$\min_a \frac{1}{2}\int_D R(k;a)^2\,dk.$$

Minimize the integrated squared residual.

Subdomain

$$\int_{D_i} R(k;a)\,dk = 0.$$

Zero average residual in each cell.

Integrals use uniform measure and Gauss–Legendre quadrature.

Three ways to compute the coefficients

Explicit fixed point: use the old policy throughout the update.

$$\widetilde{c}_{m+1,i} = \frac{c_m(g_m(k_i))}{\sqrt{\beta F'(g_m(k_i))}}, \qquad g_m(k) = F(k) - c_m(k).$$

Time iteration: hold tomorrow's policy fixed and solve for today's z_i.

$$z_i = \frac{c_m(F(k_i) - z_i)}{\sqrt{\beta F'(F(k_i) - z_i)}}.$$

Solve by bisection at each node, then refit coefficients.

Damped Newton: solve the projection equations $H(a) = 0$.

$$J_H(a_m)\Delta a = -H(a_m), \qquad a_{m+1} = a_m + \lambda_m \Delta a.$$

Fixed point and time iteration use collocation in this experiment. Least squares uses Gauss–Newton. The line search maintains feasibility and decreases the objective.

From policy updates to new coefficients

Start every solve from the same consumption policy:

$$c_0(k) = \frac{1}{6} + 0.08(k-1) = \frac{1}{6} + 0.016x(k).$$

1. Construct $c_m(k) = \sum_{j=0}^{n-1} a_j^{(m)} \phi_j(k)$.
2. Evaluate the explicit update or solve the time-iteration equation at each k_i.
3. Fit the updated values:

$$Ba^{(m+1)} = \widetilde{c}_{m+1}, \qquad B_{ij} = \phi_j(k_i).$$

4. Repeat and check the collocation residual.

For the polynomial bases, $a^{(0)} = (1/6, 0.016, 0, \ldots)$. For linear hats, initialize $a_j^{(0)} = c_0(k_j)$; at the knots, $B = I$.

The optimal policy moves capital toward one

k	$c(k)$	$g(k)$
0.80	0.151133	0.806490
0.90	0.159081	0.903253
1.00	0.166667	1.000000
1.10	0.173946	1.096740
1.20	0.180961	1.193478

Consumption rises with capital.

$$k < 1: \quad g(k) > k,$$

$$k > 1: \quad g(k) < k.$$

Saving moves capital toward $k^* = 1$.

Solver comparison: Newton is faster here

Chebyshev basis, $n = 9$, collocation.

Solver	Steps	Time (ms)	Max Euler	Policy error
Time iteration	193	10.016	8.96×10^{-12}	1.31×10^{-11}
Explicit fixed point	179	1.335	8.93×10^{-12}	1.31×10^{-11}
Newton	**4**	**0.131**	8.56×10^{-12}	1.33×10^{-11}

Nearly identical accuracy; Newton is about $10\times$ faster than fixed point and $76\times$ faster than time iteration in these runs.

Projection rankings depend on the error measure

Chebyshev, $n = 5$ Newton / Gauss–Newton

Projection	Max Euler	RMS Euler	Policy error
Collocation	1.33×10^{-7}	7.57×10^{-8}	2.09×10^{-7}
Galerkin	2.45×10^{-7}	6.62×10^{-8}	1.46×10^{-7}
Least squares	2.41×10^{-7}	6.62×10^{-8}	1.48×10^{-7}
Subdomain	4.18×10^{-7}	8.24×10^{-8}	1.12×10^{-7}

Linear hats, $n = 9$

Projection	Max Euler	RMS Euler	Policy error
Collocation	5.69×10^{-5}	2.06×10^{-5}	5.58×10^{-5}
Galerkin	3.76×10^{-5}	1.08×10^{-5}	1.34×10^{-5}
Least squares	3.57×10^{-5}	1.01×10^{-5}	1.09×10^{-5}
Subdomain	3.97×10^{-5}	1.02×10^{-5}	8.34×10^{-6}

Takeaways

- **Separate the numerical choices.** The basis represents the policy, the projection defines the conditions, and the solver finds the coefficients.
- **Evaluate away from fitting nodes.** Report maximum Euler residual, RMS residual, and consumption-policy error.
- **For this smooth problem:** Chebyshev collocation with damped Newton and about 9–13 coefficients offers speed and accuracy.

Fast convergence does not guarantee an accurate approximation.

Appendix: basis accuracy and conditioning

Accuracy with $n = 9$, collocation and Newton

Basis	Max Euler	Policy error	Time (ms)
Chebyshev	8.56×10^{-12}	1.33×10^{-11}	0.131
Legendre	8.57×10^{-12}	1.33×10^{-11}	0.137
Monomial	8.57×10^{-12}	1.34×10^{-11}	0.140
Linear hats	5.69×10^{-5}	5.58×10^{-5}	0.122

Condition number of the collocation basis matrix B

Basis	$n = 5$	$n = 9$	$n = 13$	$n = 21$
Chebyshev	1.41	1.41	1.41	1.41
Legendre	3.12	4.71	5.96	7.97
Monomial	19.6	627	2.06×10^{4}	2.31×10^{7}
Linear hats	1.00	1.00	1.00	1.00

Source: numerical_methods(2).pdf, Tables 2 and 5, pp. 5 and 8. Reported results; no new timing runs.

Appendix: numerical accuracy checks

Errors use 2,001 interval midpoints and both endpoints, including points outside the fitting set.

Reference / robustness check	Reported value
Maximum scaled Euler residual of the reference	1.17×10^{-15}
Consumption change: 25 to 29 coefficients	1.03×10^{-15}
Consumption change: use time iteration	1.49×10^{-12}
Difference from independent 600-period Euler system	2.50×10^{-16}
Largest change when doubling quadrature nodes ($n = 9$)	3.24×10^{-7}

The independent Euler-system comparison uses five initial capital levels. The largest quadrature change is for linear hats with subdomain projection and remains below its policy error. These are numerical checks, not a formal global error bound.

Source: numerical_methods(2).pdf, Accuracy checks and Table 6, pp. 8–9. Reported results; no new timing runs.